

MS5612 Assignment 2: Valuing Option to Abandon

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1 Problem setup

The project has a current value $S_0 = 500$ and a life of $t = 5$ years, modeled with a 5-step binomial tree. The risk-free rate is $r_f = 5\%$ per annum. The up and down factors are

$$u = 1.649, \quad d = 0.607.$$

At the end of each year the firm may abandon the project and collect a fixed salvage value instead of continuing:

Year	Abandonment value
1	300
2	210
3	147
4	103
5	72

We want the value of the abandonment option, i.e. how much more the project is worth because management has this flexibility, compared to a project with no abandonment right at all.

2 Risk-neutral probability and discounting

The one-period risk-neutral (up) probability is

$$p = \frac{(1 + r_f) - d}{u - d} = \frac{(1 + 0.05) - 0.607}{1.649 - 0.607} = \frac{1.05 - 0.607}{1.042} = 0.4251,$$

and

$$q = 1 - p = 0.5749.$$

Each year's expected value is discounted with a simple discount factor

$$DF = \frac{1}{1 + r_f} = \frac{1}{1.05} = 0.9524.$$

3 Underlying project value tree

The underlying (project value with no flexibility) evolves as $S_{n,k} = S_0 u^k d^{n-k}$, where n is the year and k is the number of up-moves. This is shown below (all figures to two decimals).

Year	Node values					
0	500.00					
1	824.50	303.50				
2	1359.60	500.47	184.22			
3	2241.98	825.28	303.79	111.82		
4	3697.03	1360.88	500.94	184.40	67.88	
5	6096.40	2244.10	826.06	304.07	111.93	41.20

4 Backward induction with the abandonment option

At each node in years 1–5, management compares the discounted expected continuation value against that year’s abandonment value and takes the larger of the two:

$$V_{n,k} = \max \left(\underbrace{\text{DF} \times [p V_{n+1,k+1} + (1-p) V_{n+1,k}]}_{\text{continuation value}}, \text{abandonment value}_n \right).$$

At year 5 (the final decision date) this reduces to $V_{5,k} = \max(S_{5,k}, 72)$. Working backward:

Year	Node values with option (bold = abandoned)					
5	6096.40	2244.10	826.06	304.07	111.93	72.00
4	3696.94	1360.86	500.90	184.39	103.00	
3	2241.30	825.15	303.72	147.00		
2	1359.14	500.35	210.00			
1	824.19	317.61				
0	507.73					

Only the “all-down” branch (the bottom node of the tree at every stage) ever falls low enough to trigger abandonment. It first becomes optimal to abandon at year 2, where the continuation value equals the salvage value of 210; along that branch the project keeps getting abandoned every year after that, since the underlying value keeps falling faster than the shrinking abandonment payouts. At year 1, the down node’s continuation value (317.61) still beats the abandonment value of 300, so the firm keeps the project open there.

5 Binomial tree diagram

Figure 1 shows the full tree. Each node reports the underlying project value S on top and the option-adjusted value V (i.e. value including the right to abandon) below it; nodes where abandonment is optimal are shaded.

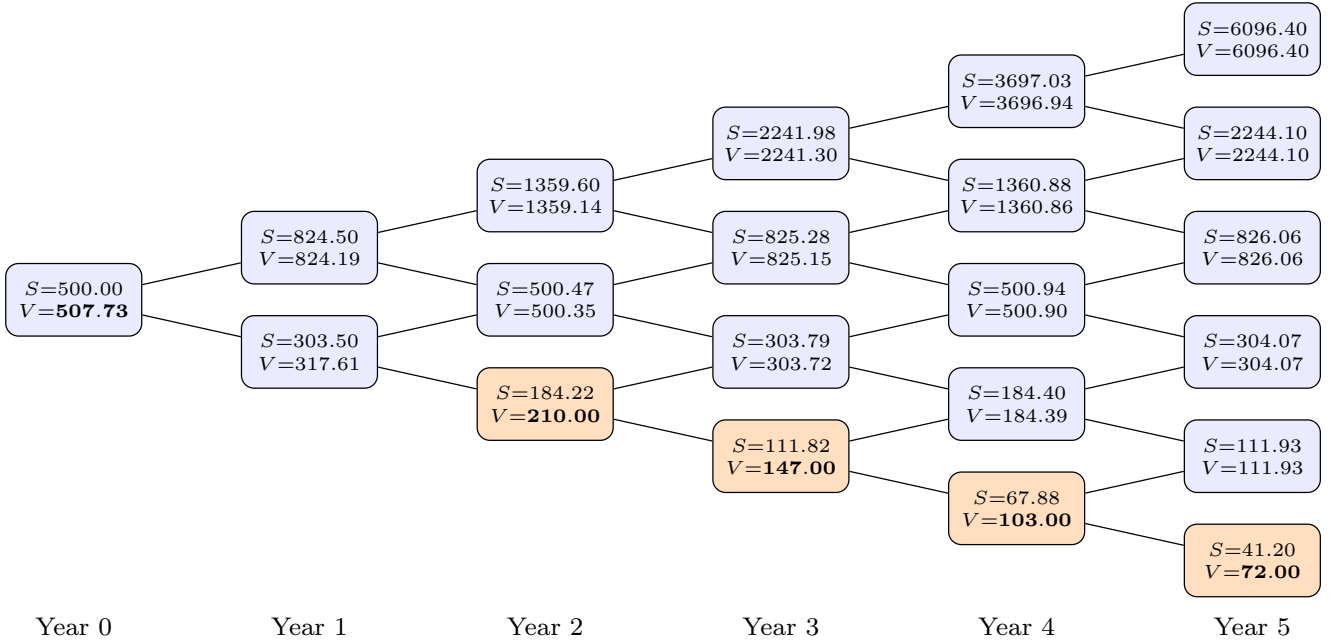


Figure 1: Binomial tree for the project value S and the option-adjusted value V . Shaded (orange) nodes are where abandonment is exercised.

6 Result

Project value with abandonment option (at $t = 0$) = 507.73

Project value with no flexibility (S_0) = 500.00

Value of the abandonment option = $507.73 - 500.00 = 7.73$

Interpretation

The abandonment option is worth relatively little here (about 1.5% of the project's value) because the project's value has to fall a long way, along the single worst-case (all-down) path, before abandoning ever beats continuing. In every other branch of the tree the project simply grows too large for the salvage values to matter, so the option only pays off in a thin slice of the outcome space.